Section 2
Cells

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COGS 17

10/10/25

Notice

MIDTERM I (125 Points)– Next Thursday!

3:30-4:50 pm (80 minute)

Exam Online

For section slides:

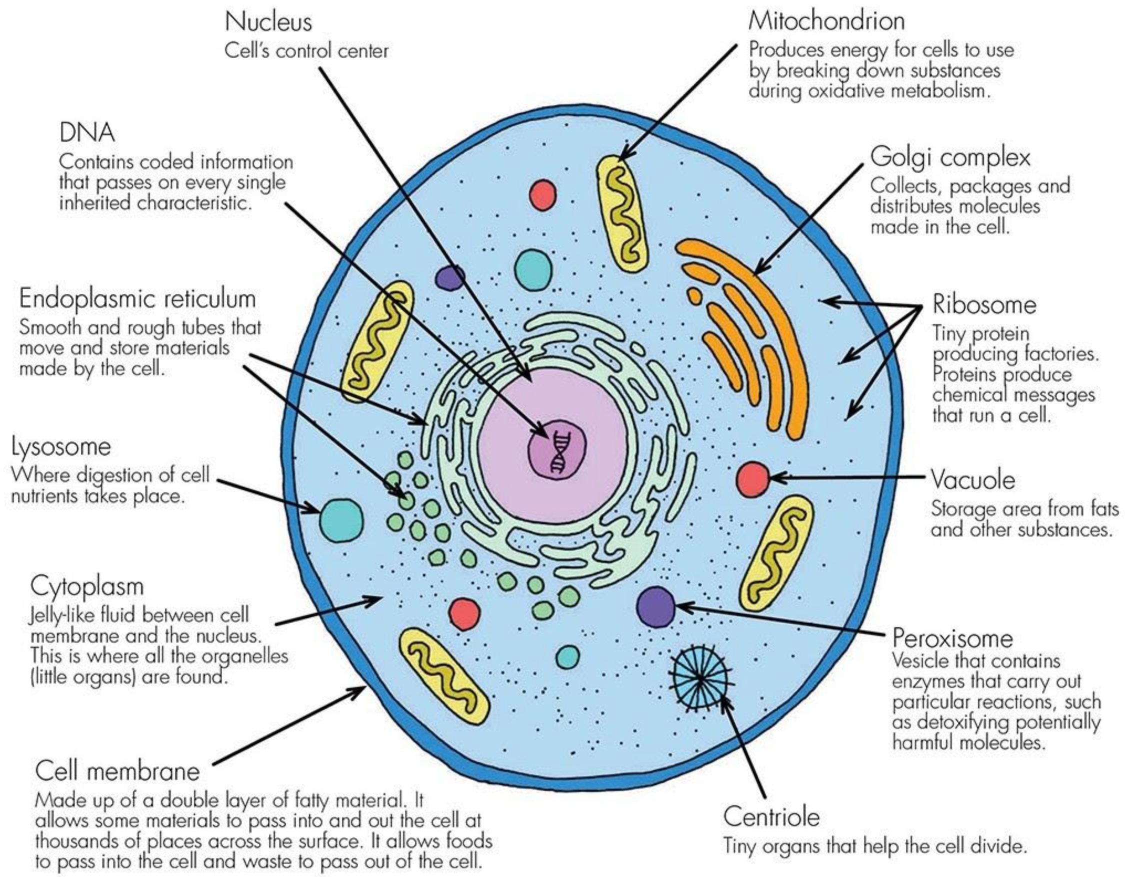
Common features of Cells

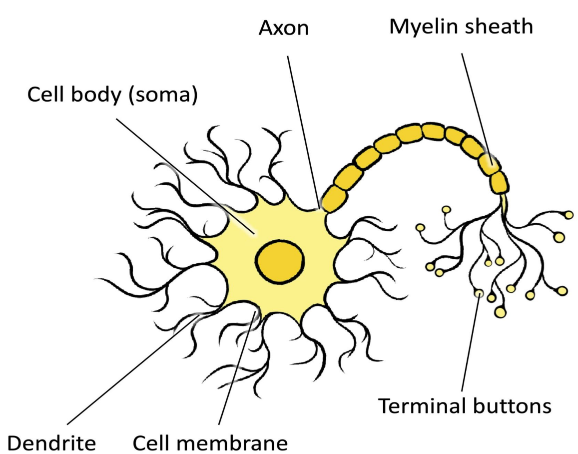
- <u>Soma</u>
  Fancy word meaning “cell body”

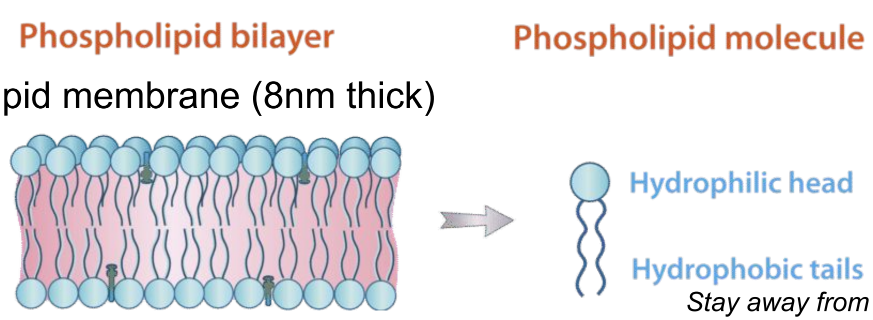
- <u>Cytoplasm</u>
  Fluid within a cell

- <u>Extracellular Fluid</u>
  Fluid outside of a cell

- <u>Cell Membrane</u>
  A double layered wall consisting of lipids (fat molecules)







Double Lipid membrane (8nm thick)

Stay away from water!

Important Organelles to Remember

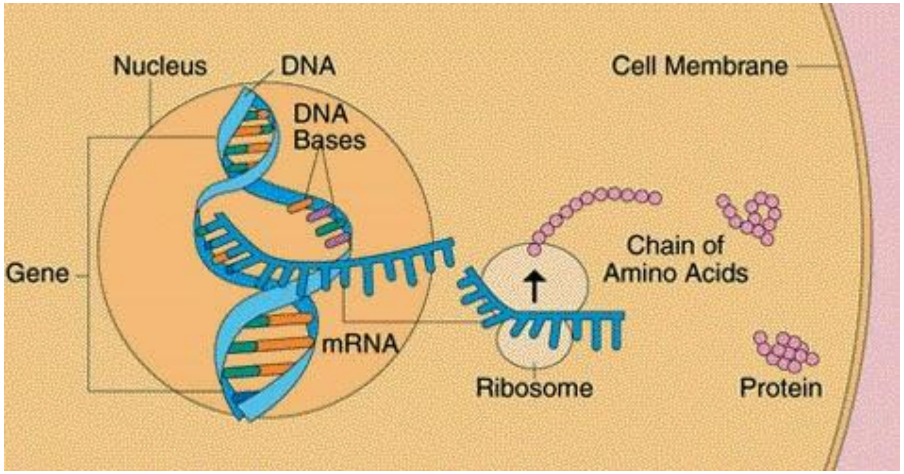
- <u>Nucleus</u> : An inner "control center" where **DNA** is stored

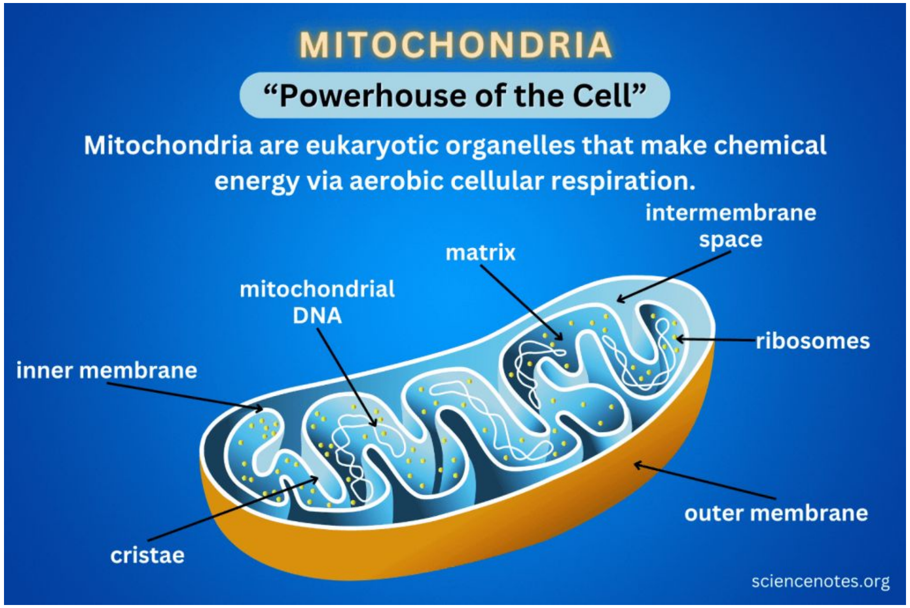
* DNA sends messenger RNA from Nucleus, mRNA attaches to Ribosomes, instructs protein production

- <u>Ribosomes</u>: Small protein producing factories

* Proteins serve many functions related to structure, transport, metabolism, membrane gates, etc…

- <u>Mitochondria</u>: The "powerhouse of the cell" , produce ATP whose breakdown frees energy to power cell functions

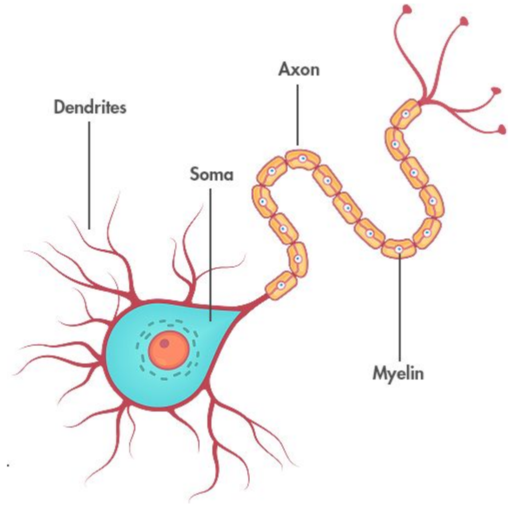




Specialized Cells of the Nervous System

2 Types of Cells

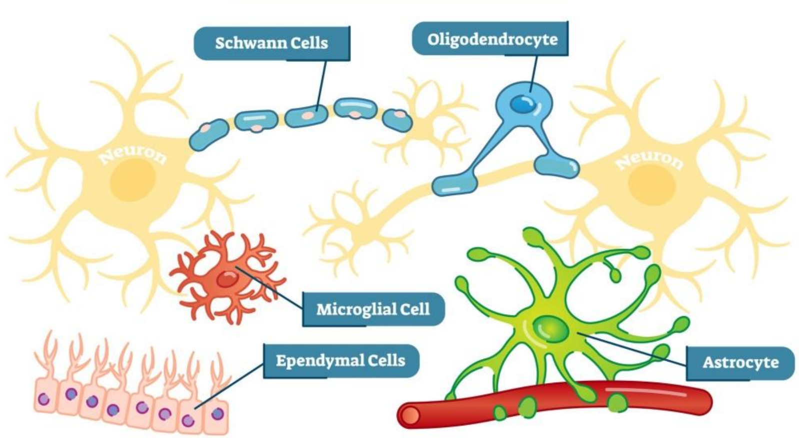
- <u>Neurons</u>
  Cells that are specialized for Information Transfer



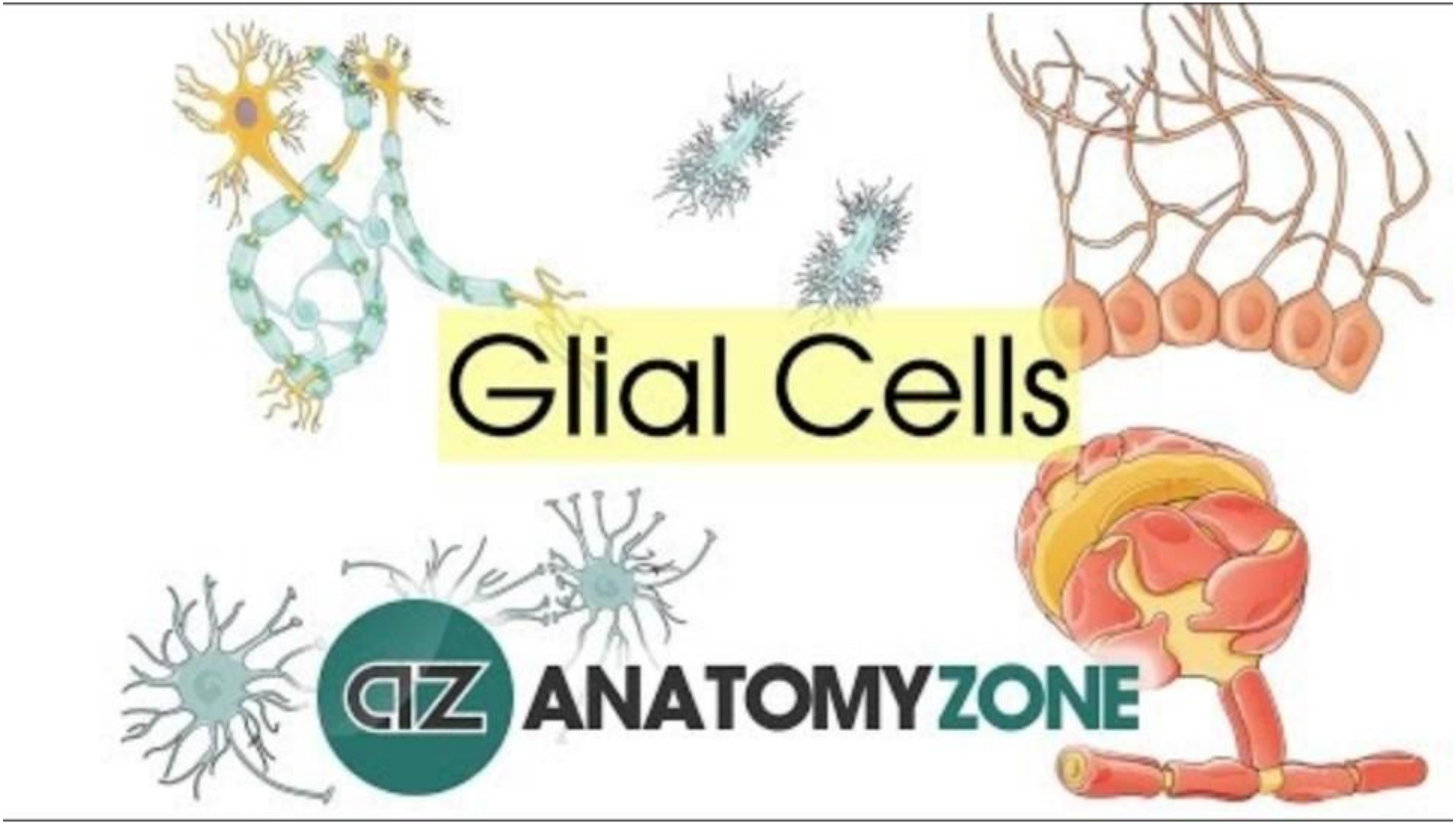
- <u>Glia Cells</u>
  Have many functions but do not participate in Information Transfer

  "Glia" meaning "Glue" which holds the nervous system together, both physically and chemically, to support Neurons

  A lot smaller than neuron (1/10 size), but greater in amount (x10 times as many), takes up 50% of brain by weight



Different Glia Cells



Different Glia Cells

- **Radial Glia**

Guide the migration and growth of neurons during fetal development

- **Ependymal Cells:**

Lines ventricles and secretes CSF into the Ventricles

- **Oligodendrocytes**

Surrounds axons in a process called myelination in the CNS

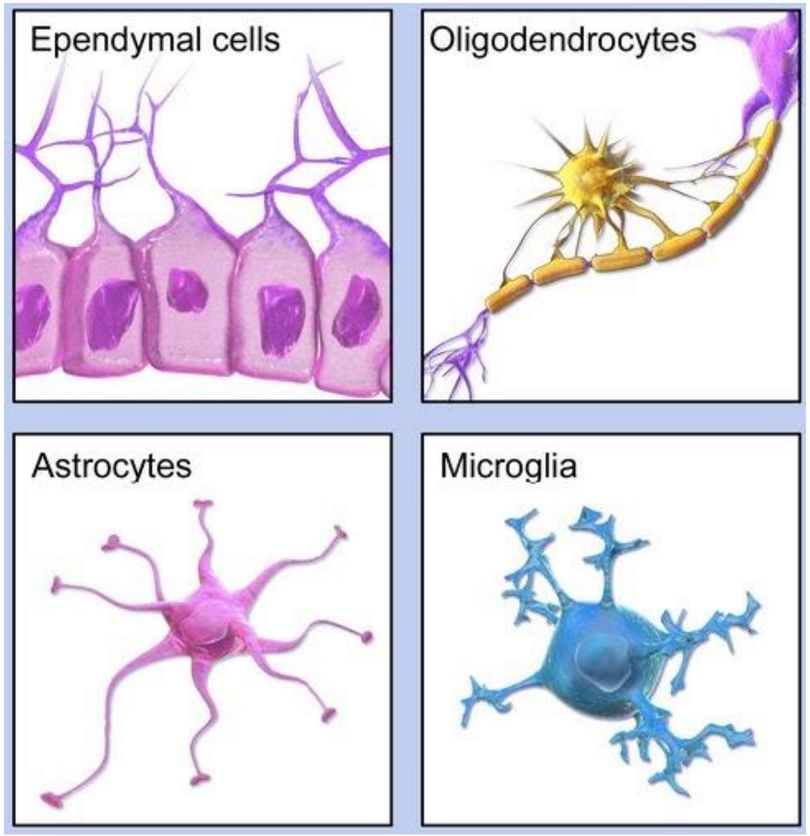
Schwann Cells: specialized Oligos which myelinate neurons of the PNS

- **Astrocytes**

Provides nutrients, recycles NTs, maintains the BBB, and numerous other functions

- **Microglia**

Removes toxins from the brain, repairs damaged neurons



Neurons

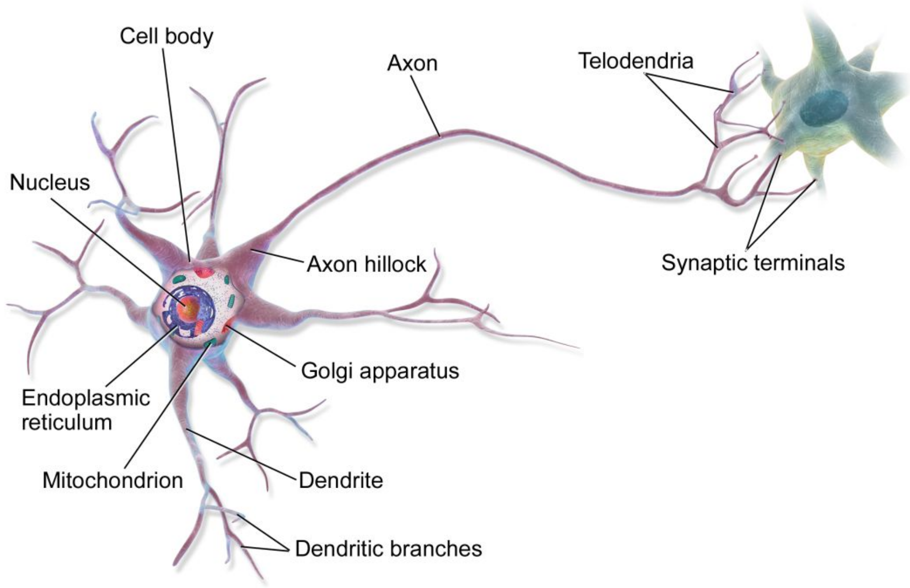
- Have very small cell-body, but branches can be 2m long

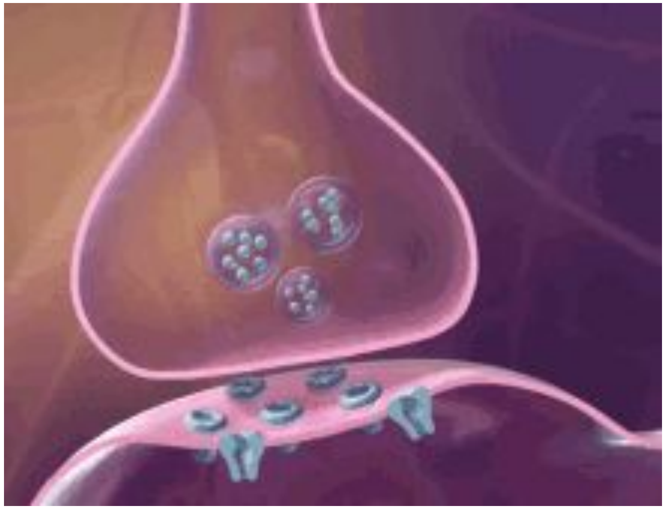
- Specialized cells for information transfer

- <u>Dendrites:</u>
    - Spiny protrusions from the Soma which receives **incoming** signals
    - Site of Postsynaptic Membranes
    - w/ receptor sites

- <u>Axons</u>
    - Long fibers which reach out to other neurons
    - Carries **outgoing** signals
    - Terminates in Presynaptic Terminals (aka. Terminal Buttons, or End Bulbs) which releases NTs into the Synaptic Cleft

- <u>Receptor Sites:</u>
  Specialized areas which interact with NTs from other neurons





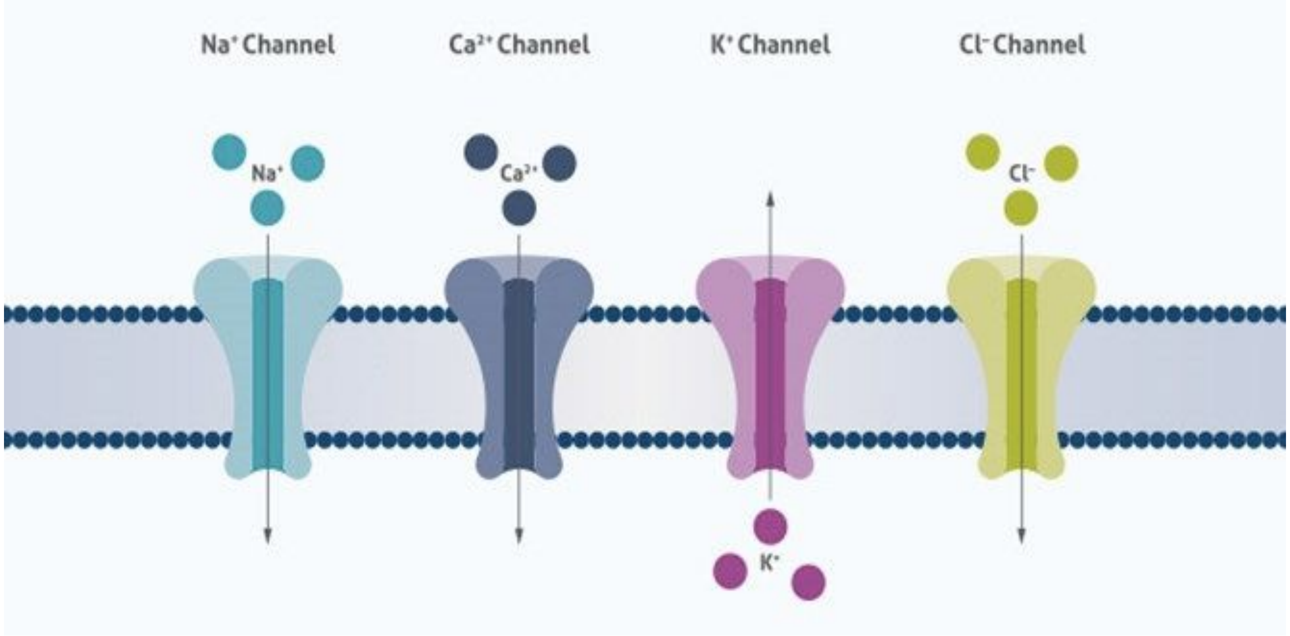
Important Concepts

How does information transfer occurs b/w neurons?

Important Ions to remember:

Na^+, K^+, Ca^{2+}, Cl^-

Sodium, Potassium, Calcium, Chloride

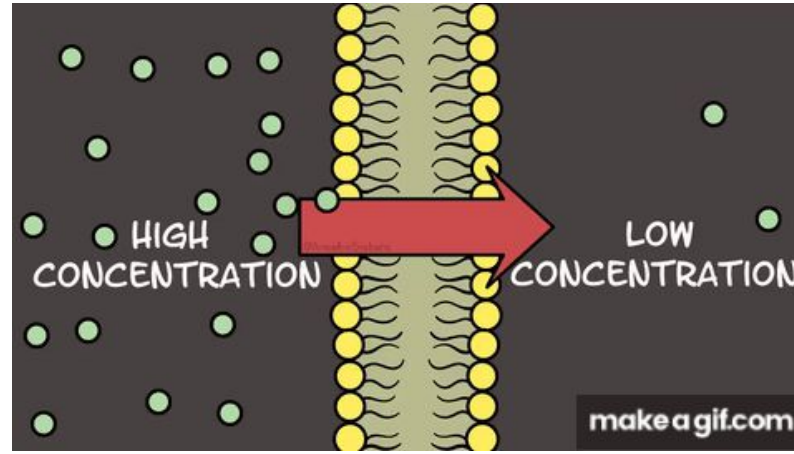


Important Concepts

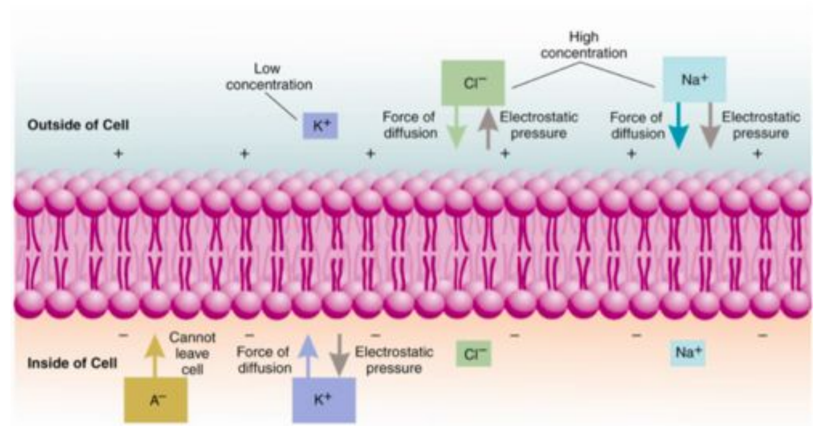
- Nature always seeks **Equilibrium**, so is neural communication (Equilibrium vs. Disequilibrium)

- How to achieve this stable state?

1. <u>Concentration Gradient:</u>
   - Molecules in areas of **greater** concentration will *diffuse* to areas of **lesser** concentration



2. <u>Electrical Gradient:</u>
   - "Identical charges REPEL", Negative repels negative charges and positive repels positive charges, but negative & positive attracts each other
     = **Electrostatic Pressure**



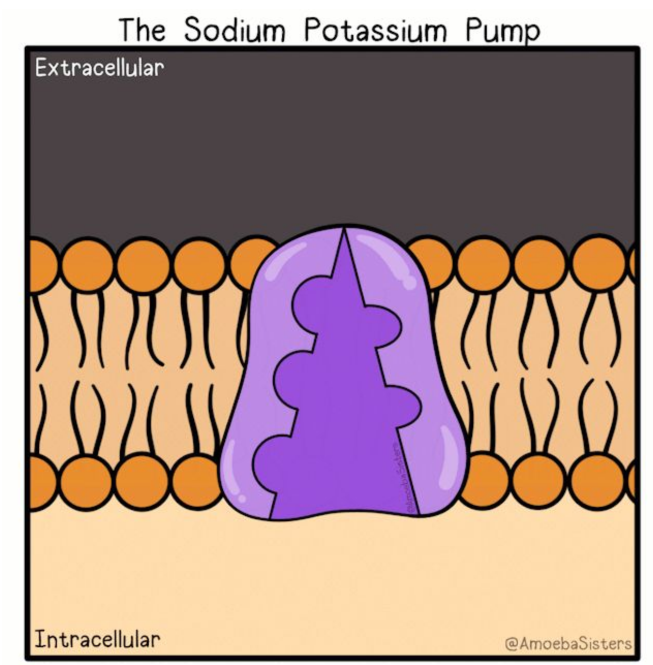
3. <u>Selective Permeability of Membranes</u>
   - Lipid bi-layers → typically impermeable to charged ions and larger molecules
   - Control which chemicals enter/leave the cell and this is done by gates that open or close to let ions pass through

Membrane Potential

- <u>Membrane Potential</u>
  - The difference in charge between the inside and outside of the cell, in milli-volts (mV)

- <u>Resting Potential:</u>
  - Typically highly polarized, **-70 mV** for Neurons (more positive outside)
  - All gates are locked (waiting for ion flows and is ready to fire)

- <u>Sodium/Potassium Pump</u>
  - Helps establish resting potential by transporting **3 Na^+ out** and **2 K^+** ions **in**



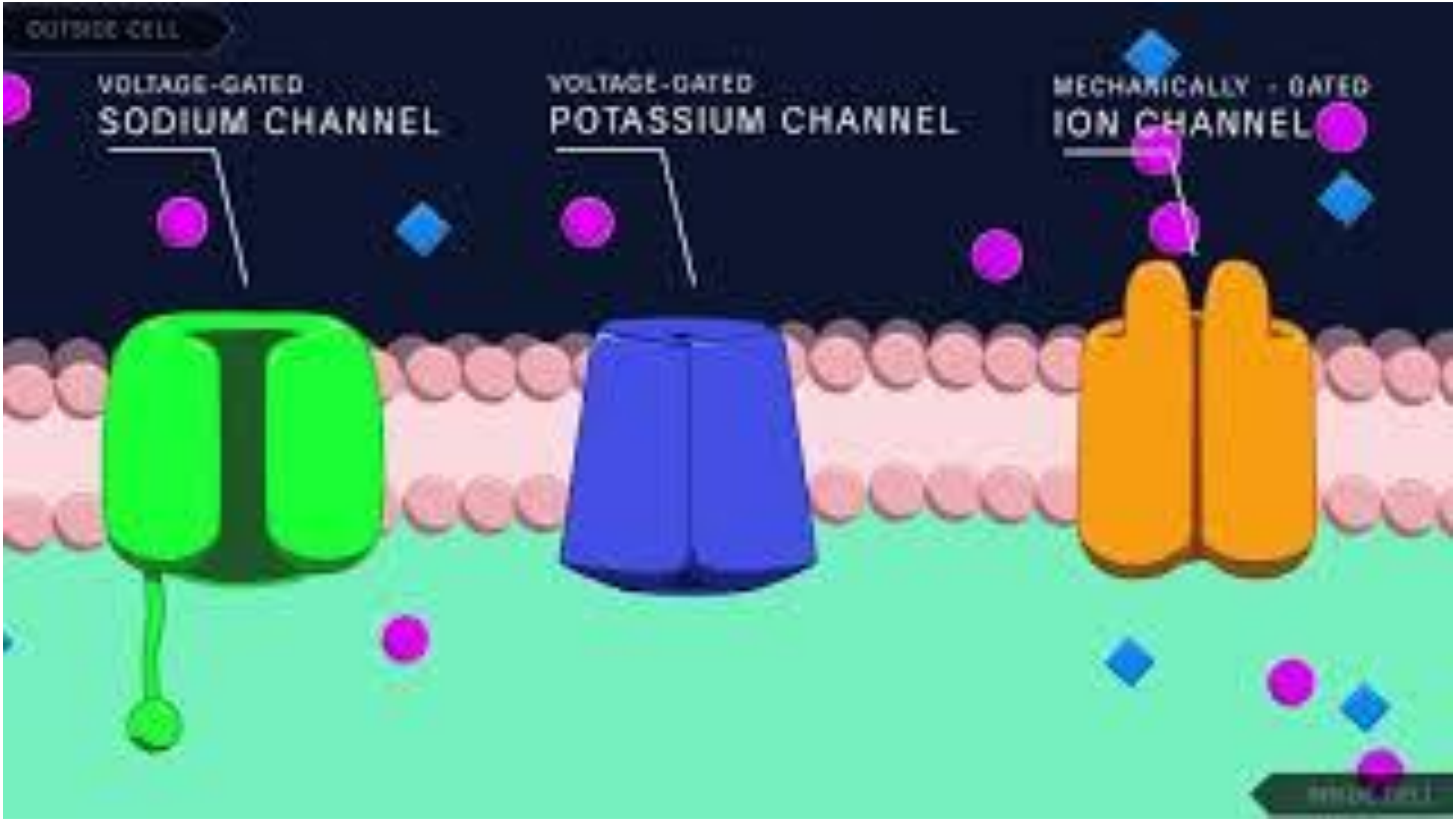
Eventually…

Na+ { Let me go inside… it's already so packed outside
waaait but the gate's closed, we're trapped!!! }

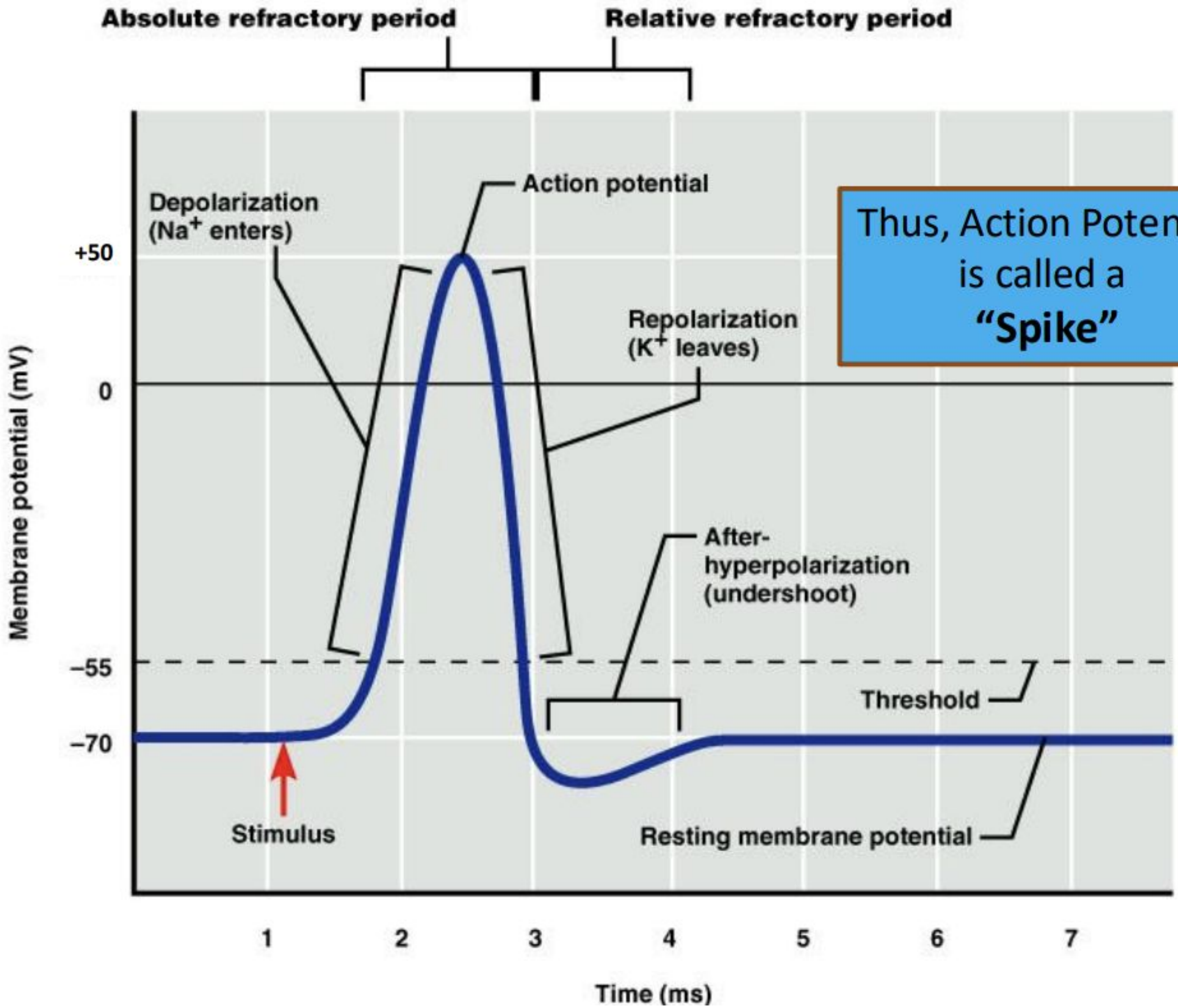
K+ { let me go outside…
waaait but outside is too positive, can't go out!!! }

→ Polarized = Large electro-chemical difference between inside/out = Ready to "fire"

Action Potential

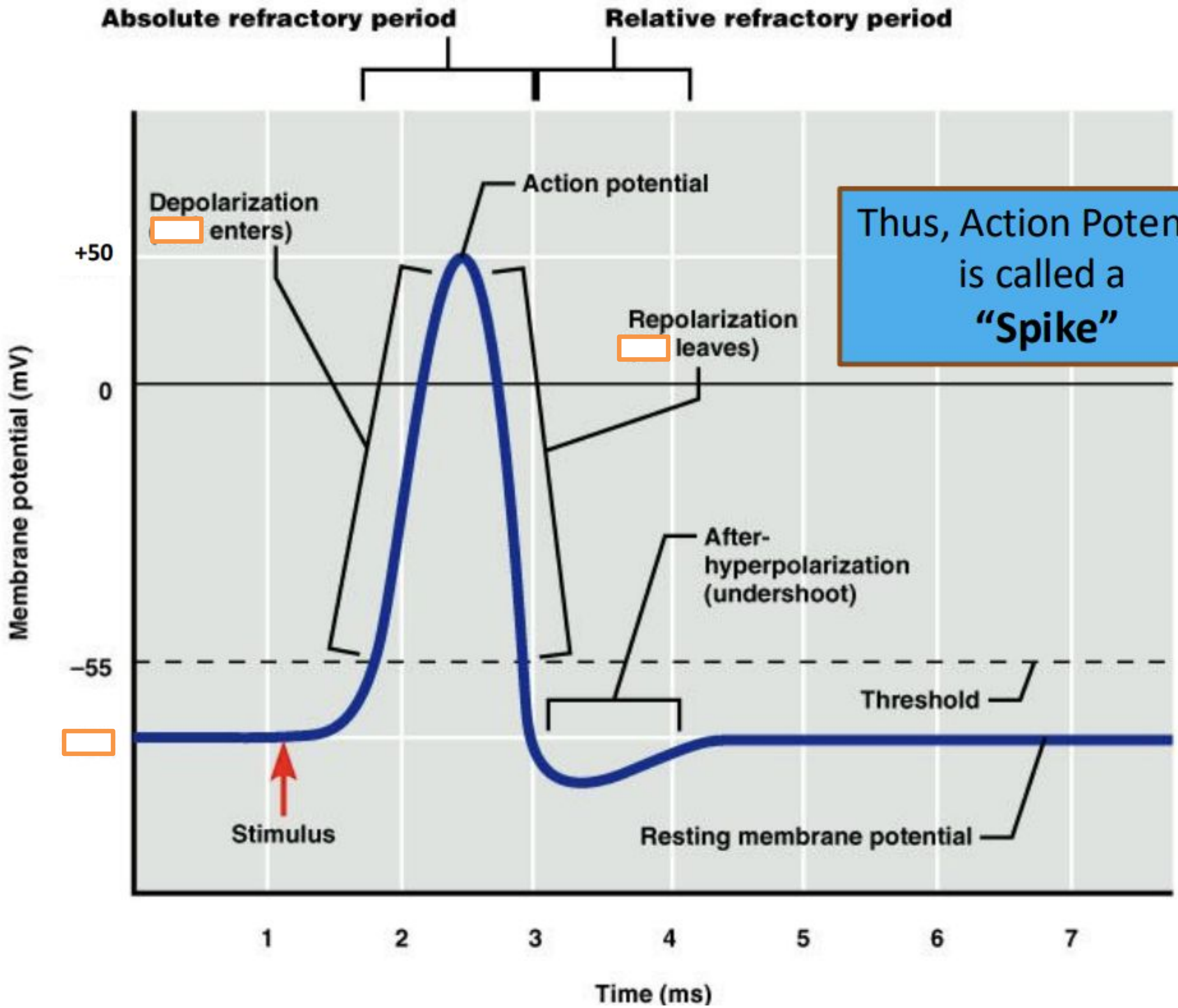


Action Potential (AP)



Thus, Action Potential is called a **"Spike"**

Action Potential (AP)



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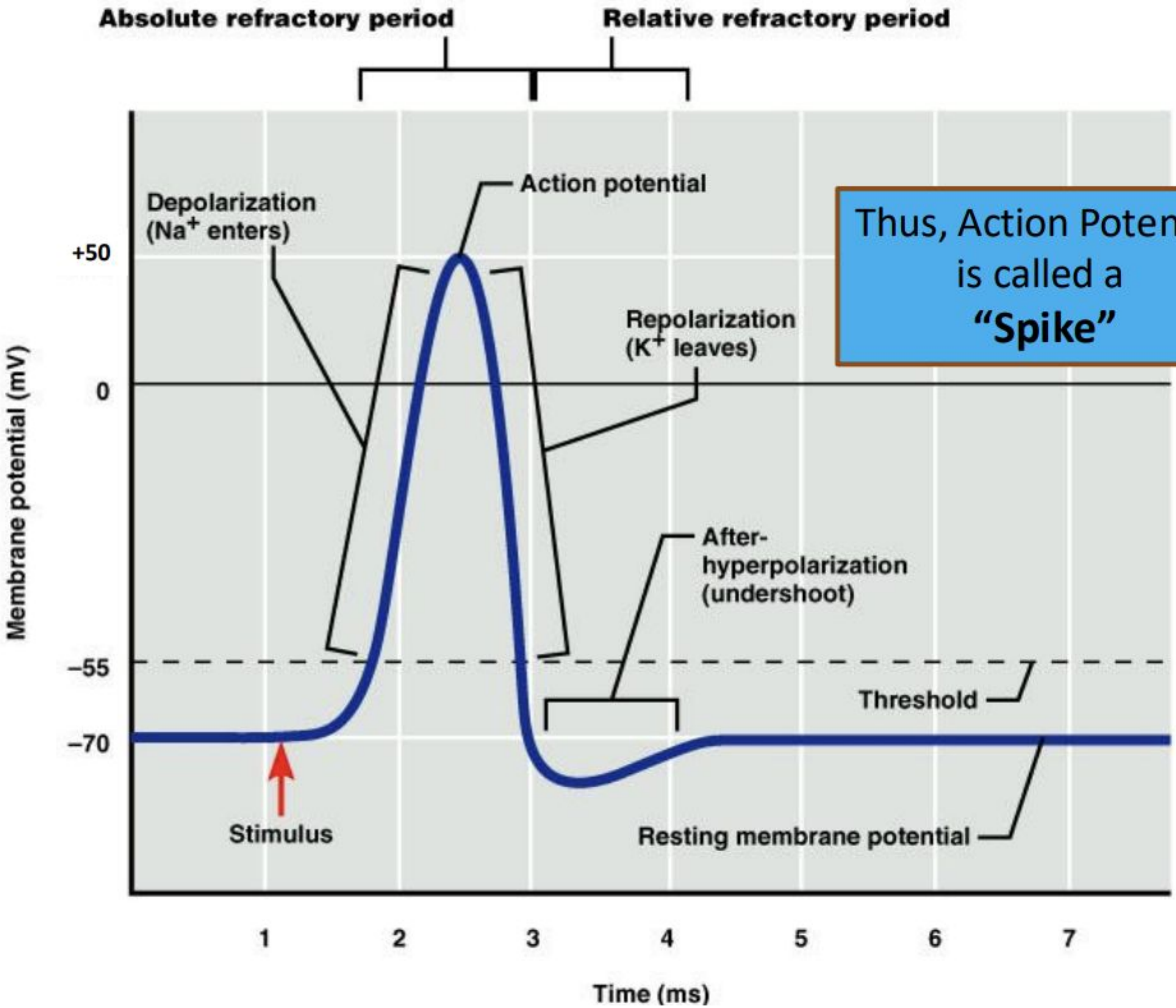
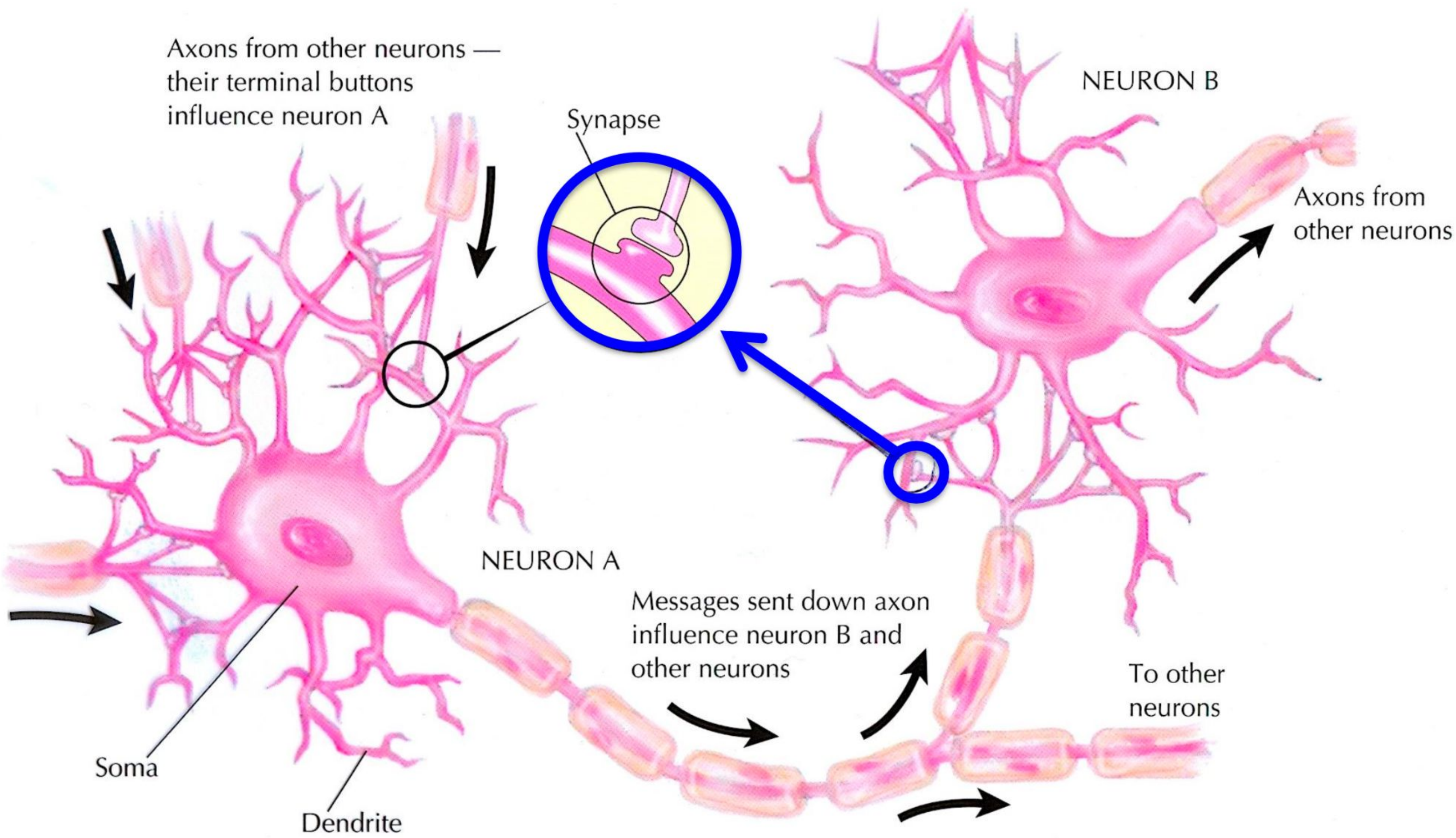


Thus, Action Potential is called a **"Spike"**

Action Potential (AP)

Propagation process

- Stimulation from Presynaptic neuron → When depolarization reaches the terminal, Ca++ enters the cell → release NTs → binds to Postsynaptic neurons → triggers AP that starts at the Axon Hillock



Action Potential (AP)

- <u>Mechanism of Action</u>
  - Na channels at axon hillock open, allowing an **influx** of **Na** ions, drastically shifting the membrane potential towards a peak of **( )** mV
  - Next Na gates open and more influx of Na ions
  - At the peak, Na channels close while K channels open, allowing an **efflux** of **K** ions, shifting the membrane potential **(positively/negatively)** to a point where it overshoots (hyperpolarizes)

  - K channels close and **Na/K pumps** start re-establishing resting potential (via **( )** Na out, **( )** K in) until membrane potential returns to -70 mV, This time period is called **Refractory Period**, during which the neuron cannot fire

- <u>All or None Law</u>
  - In a given cell, AP will always have the same amplitude and velocity regardless of the intensity of the stimulus that triggered it

Action Potential (AP)

- Mechanism of Action
    - Na channels at axon hillock open, allowing an **influx** of **Na** ions, drastically shifting the membrane potential towards a peak of +50 mV
    - Next Na gates open and more influx of Na ions
    - At the peak, Na channels close while K channels open, allowing an **efflux** of **K** ions, shifting the membrane potential negatively to a point where it overshoots (hyperpolarizes)

    - K channels close and **Na/K pumps** start re-establishing resting potential (via 3 Na out, 2 K in) until membrane potential returns to -70 mV, This time period is called **Refractory Period**, during which the neuron cannot fire

- All or None Law
    - In a given cell, AP will always have the same amplitude and velocity regardless of the intensity of the stimulus that triggered it

Graded Potentials (GP)

- Not all Neurons show AP
- Cells that fire GP may release MORE or LESS neurotransmitter (e.g., hair cells = auditory receptors)

- GP vs AP
  - GPs can vary in amplitude while APs do not
  - GPs do not have refractory periods while APs do

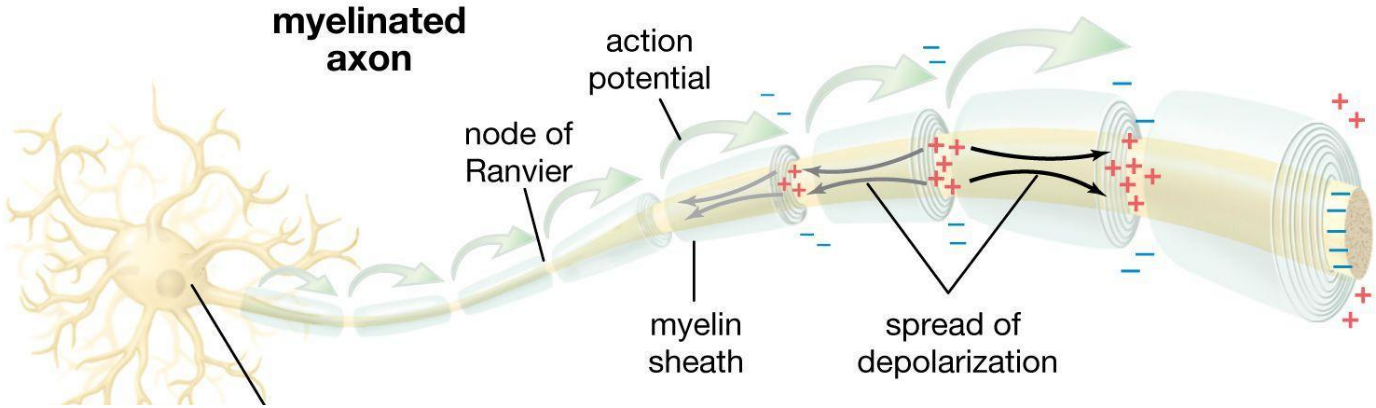
Action Potentials	Graded Potentials
Travel very long distances (meters)	Only travel very short distances (mm)
All the same size ("all or none")	Many different sizes
Rapid positive voltage deflection (depolarization)	Small positive or negative voltage deflection (EPSP or IPSP)
Maximum firing rate about 500/sec	Can be made quickly, even simultaneously
Can regenerate themselves	Decrement (steadily decrease) in space & time
Travel in only one direction in the axon	Propagate in all directions

Myelination

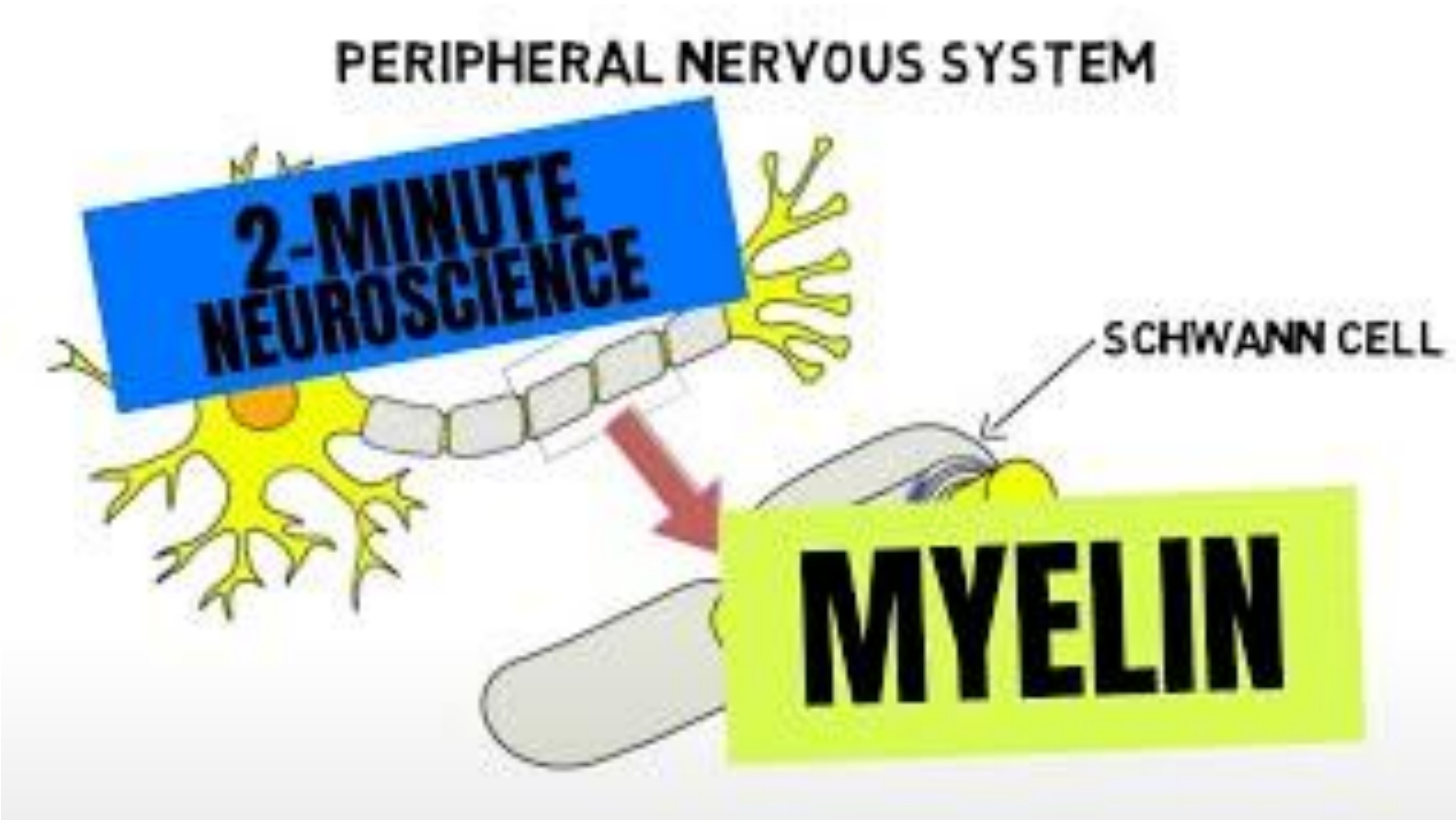
- **Speed up** AP

- Glia cells are wrapped around the axon, w/ gaps called "Nodes of Ranvier", act as an insulator
- **Electrical conduction** (electricity flows thru axon insulation) in myelinated portions: very fast but decays over time → Reboost to original strength occurs at…

- <u>Nodes of Ranvier:</u>
    - The small gaps between myelin sheaths
    - Here, electrical signal triggers the slower but stronger Ionic Conduction, which in turn triggers Electrical Conduction that moves rapidly under next sheath to next node, etc, etc



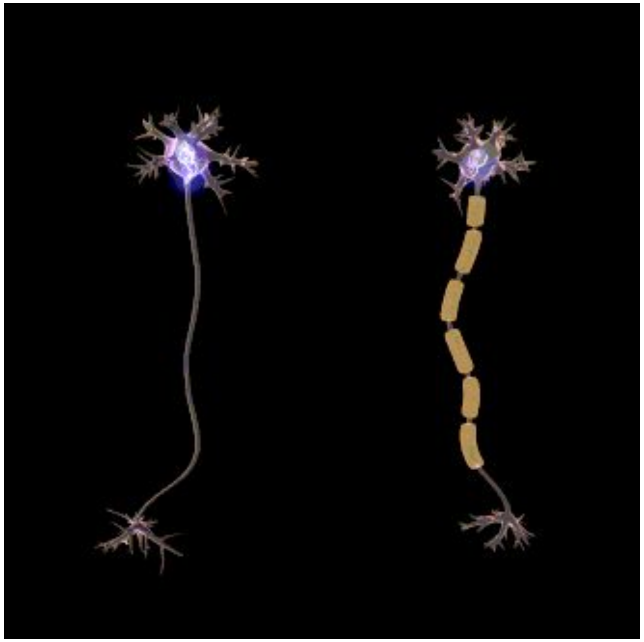
Myelination

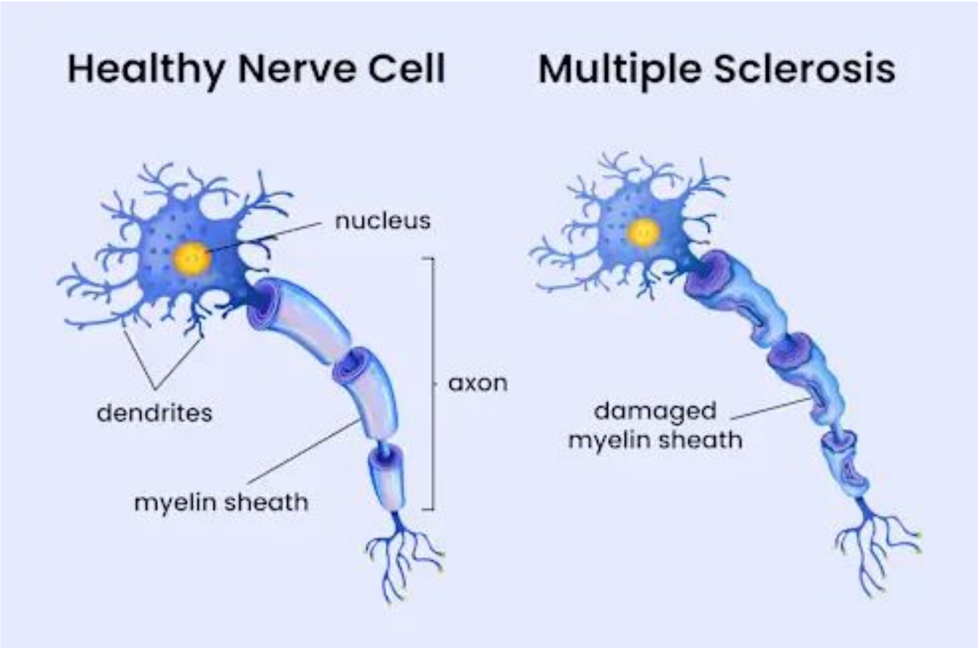


Myelination

- <u>Saltatory Conduction:</u>
  - Nerve impulse “jumps” from one node to another in a myelinated cell
  - Increases overall speed of impulse

- <u>Multiple Sclerosis (MS):</u>
  - A neurodegenerative disease where myelin degrades over time
  - Electrical signals decay quickly and AP fail





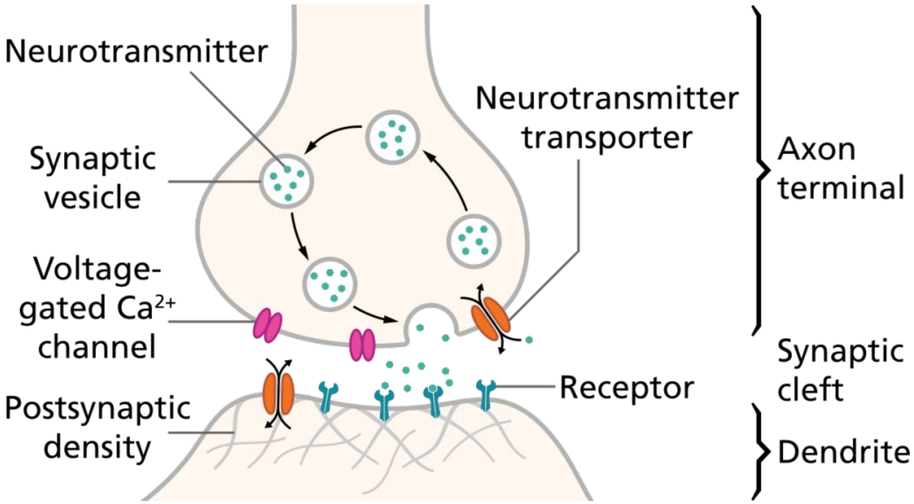
The Synapse

- Presynaptic cell + Synaptic Cleft + Postsynaptic cell = The Synapse

- Presynaptic cells release NTs into the cleft via **Exocytosis (releasing the NTs)**
    - NTs are packaged into vesicles

- **Influx of Ca** initiates the exocytosis
    - Ca opens the Fusion Pore which **binds vesicles** to the presynaptic cellular membrane

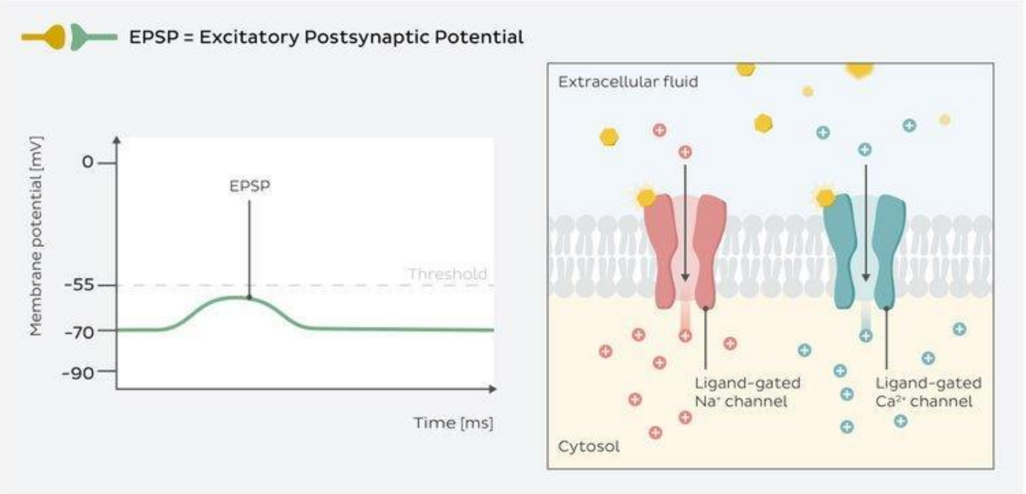
- Following exocytosis, **NTs** passively **diffuse** across the synaptic cleft and binds to NT-specific receptor sites on postsynaptic neurons

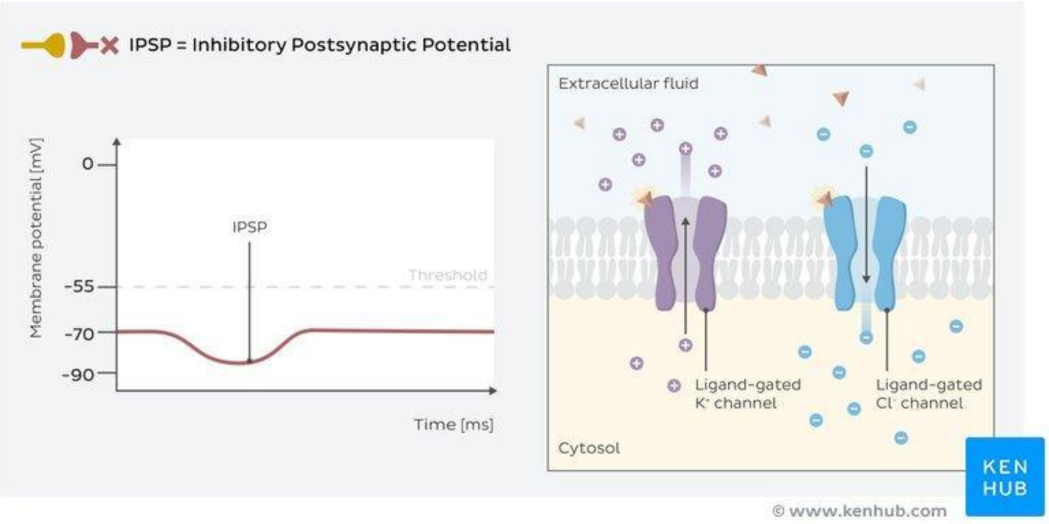


Polarity of Postsynaptic Cells

- **EPSP**
  - Increases a cell's likelihood of releasing NTs, **more** likely to "fire"
  - Usually due to Na+ entering the cell

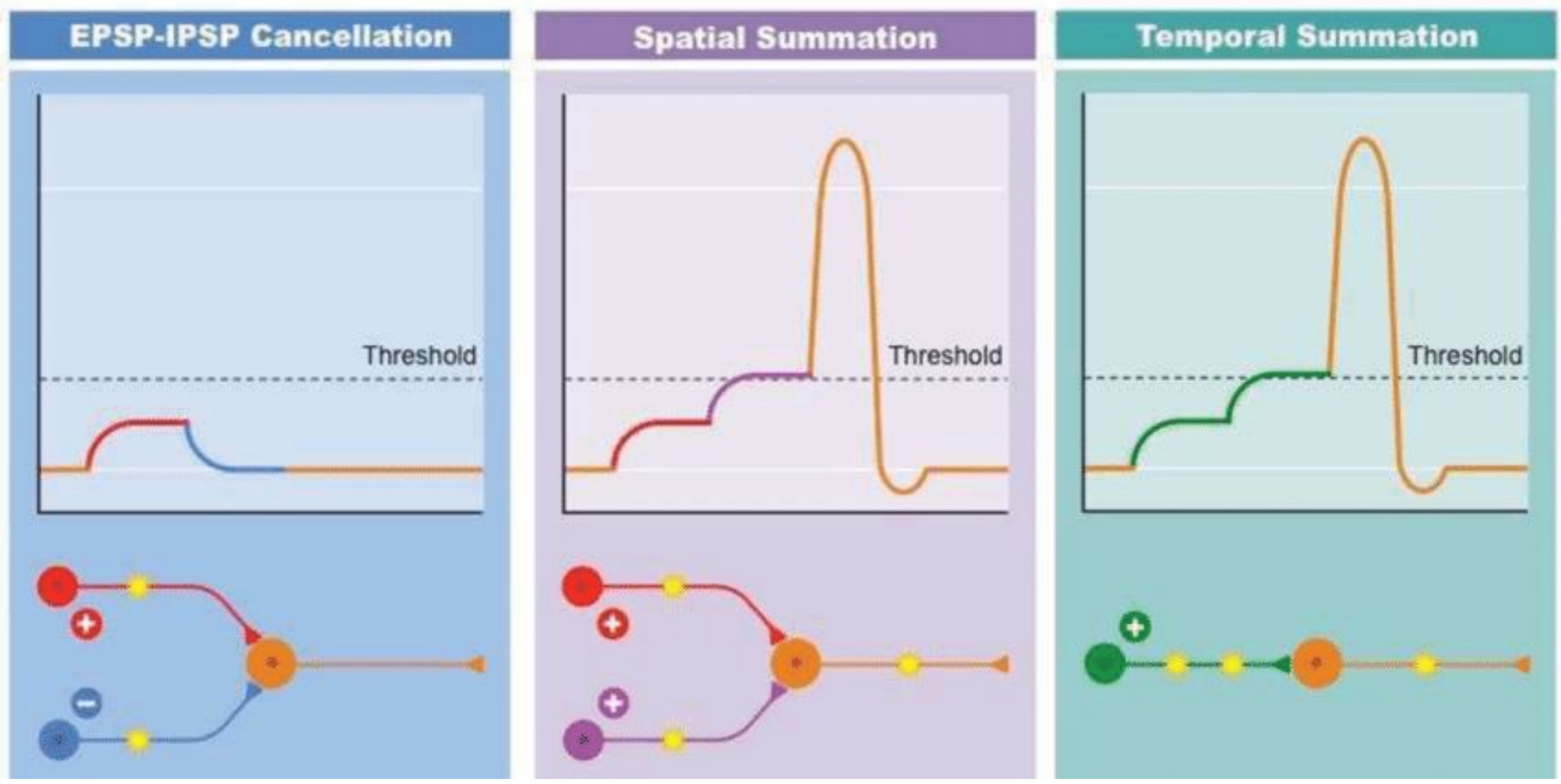
- **IPSP**
  - Decreases a cell's likelihood of releasing NTs, **less** likely to "fire"
  - Usually due to K+ entering or Cl- exiting



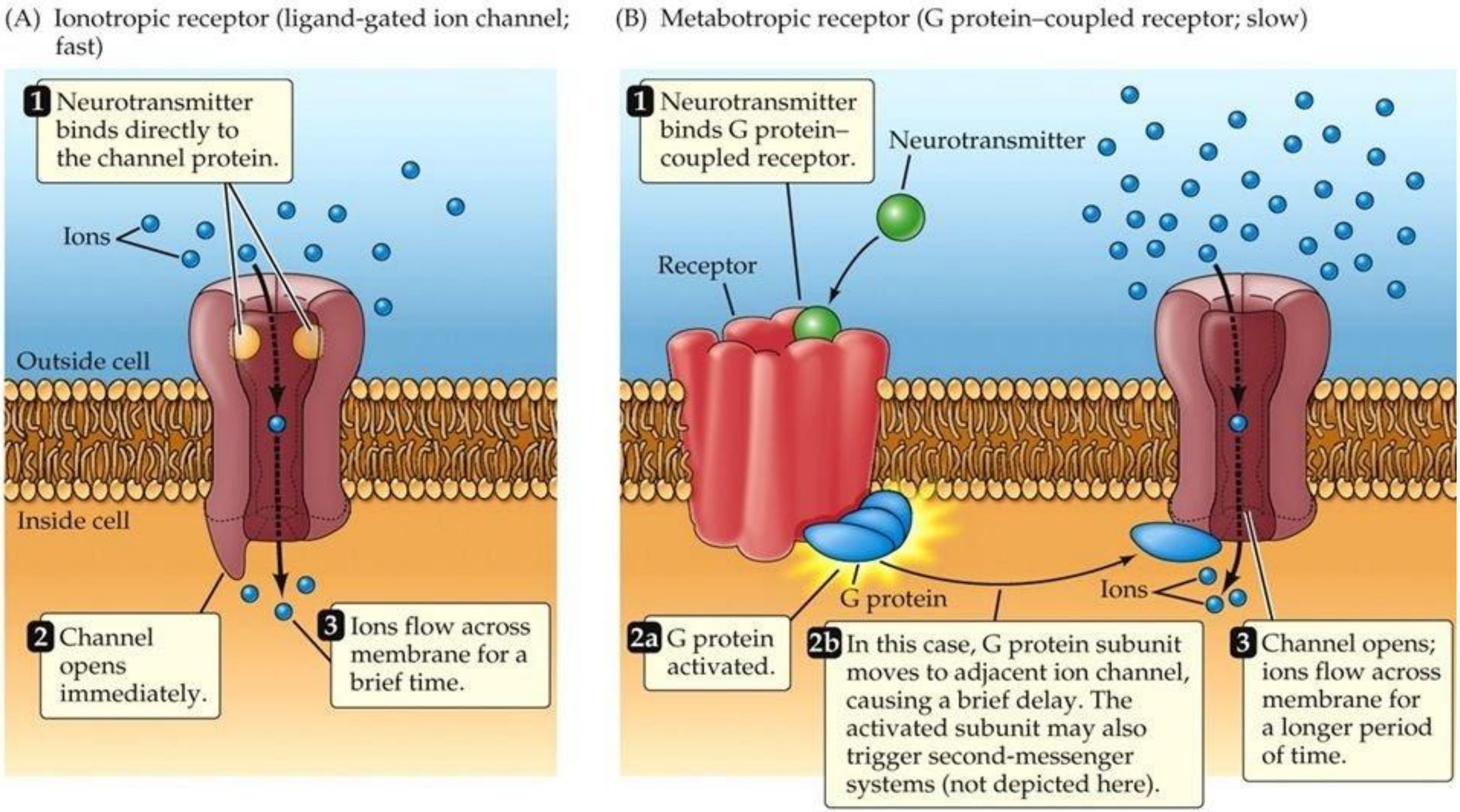


Polarity of Postsynaptic Cells

- Summation
    - A neuron’s response = sum of EPSPs and IPSPs
    - **Temporal Summation:** one or more cells **repeatedly** stimulate another in rapid succession
    - **Spatial Summation:** multiple cells converge on a single **location** on a cell at the same time



Mechanisms of Neurotransmitters



Ionotropic

- Directly affects ion gates
- Rapid and Short-lived responses
- Best for sending info about changing inputs

Metabotropic

- Causes metabolic changes in Postsynaptic cell
- Activation of G protein and second messenger
- Slower but long-lasting responses

Some Neurotransmitters and their Functions

Chemicals are called NTs if they impact nearby neurons

Neurotransmitter	Functions
Acetycholine (Ach)	•All neuro-muscular junctions •Cortical arousal
GABA	•Most common inhibitory NT •Regulate anxiety
Glutamate	•Most common excitatory NT •Learning •Perception •Schizophrenia
Serotonin (5HT)	•Often acts as a neuromodulator •Mood regulation, sleep, perception
Dopamine	•Reinforcement •Attention •Motor control
Norepinephrine	•Arousal •Attention
Epinephrine (adrenalin)	•Arousal •Attention
Substance P	•Pain (damage, itch, extreme temperatures, etc)
Endorphins	•Counter effects of Substance P
Hormones	•Testosterone, estrogen, cortisol, oxytocin, endorphins, etc

Agonist vs Antagonist

- **Agonists**: **Increases** the effect of a NT
- **Antagonist**: **Decreases** the effect of a NT

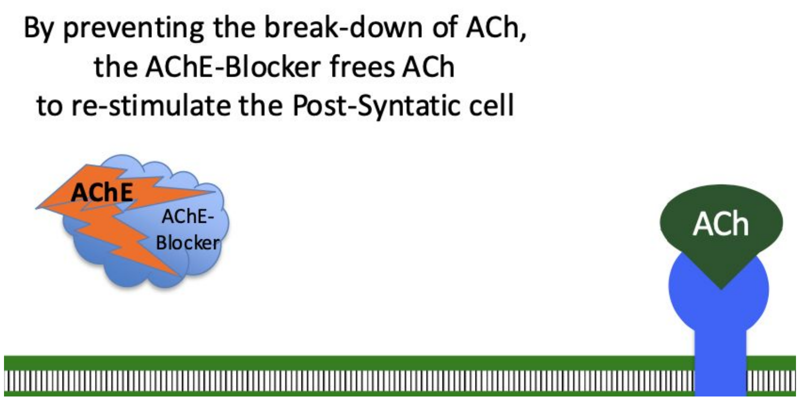
What works as an Agonist vs. Antagonist often depends on how NT is typically processed in the cleft

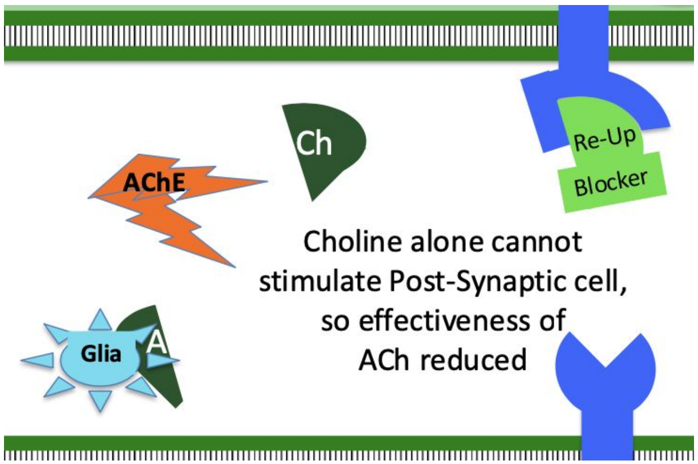
Some examples:

- Acetylcholinesterase: Enzyme in the cleft which breaks down ACh into…
    - Acetate (usually carried off by Glia cell)
    - Choline (can be re-uptaken into the pre-synpatic cell for re-use).

AchE-blocker = ACh Agonist, prevents breakdown of ACh, frees ACh to re-stimulte the Post-synaptic cell

Reuptake-Blocker = ACh Antagonist, Choline alone cannot stimulate Post-synaptic cell → effectiveness of ACh is reduced





Other Factors affecting Function

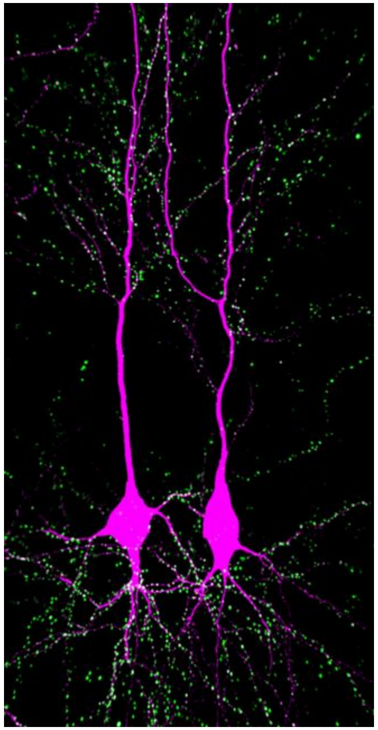
1. (Gene transcription) Activation of DNA sequences initiated the production of proteins for structural and chemical changes within cell

2. (Dendritization) Repeated activity leads to more dendritic spines and more receptor sites

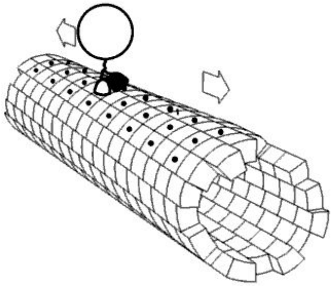
3. Receptor Sites can be blocked by NT mimics that do not readily detach
   - E.g., LSD binds to Serotonin sites

4. Some NTs may require Hours/Days to replenish
   - Carried by Kinesin molecules (walk along micro-tubules from soma to terminal to transport NT)

5. Some NT precursors can pass the BBB and be used as medication (e.g., L-DOPA, a dopamine precursor)



dendritic tree of two hippocampal pyramidal neurons



Other Factors affecting Function

<u>Exceptions: Receptor Sites on **PRE-synaptic** Terminal</u>

- Auto-Receptors
    - Some axons have receptor sites for their own NT (usually inhibitory)
    - This acts as a negative feedback loop which prevents NT release if there is already a lot of the specific NT in the cleft

- Axoaxonic Synapses (Axon to Axon)
    - Presynaptic Terminal may have Receptor Sites for Inhibitory or Excitatory NT from another cell
    - Brain Endorphins stimulate opiate receptors on terminal of pain cell, inhibiting release of Substance P